

SA2 Mock Examination — Set 2

Lectures 17–29 · Exam 2 / FA-SA2 Scope

77 Questions · Single Best Answer · Weighted by Official Lecture Hours · Fully New Item Set

This is a second, independent 77-question SA2 mock exam covering Lectures 17–29, built to the same official lecture-hour weighting as the first SA2 mock exam — but with **no question overlap** with that exam. If you've already worked through Mock Exam 1, this set gives you a genuinely fresh pass at the same material.

Sourcing. Most items are pulled from the remaining, previously-unused portion of the fully re-verified 225-question Lec17–29 past-paper key. A few archive items originally flagged as ambiguous (Q129, Q138, Q160, Q165) are used here too, each resolved to a single best answer using standard physiology, with the reasoning stated plainly. Lecture 29 again required the most supplementation — its archive pool was exhausted by Mock Exam 1 — so this set draws on previously-unused vignette/pharmacology items from “Mixed / Other Topics,” plus five original questions (†) on spinal cord syndromes, peripheral neuropathy, sensory exam technique, and NIHSS scoring. Lecture 27 similarly ran out of archive items, filled with three originals on reflex grading and pathological reflexes; Lectures 22 and 26 each add two or three originals on content the archive never directly tested.

Every question in the Answer Key states exactly where it came from — verified, verified-and-resolved, reclassified, or original (†) — so you can weight your confidence accordingly.

How to use this booklet

1. Work through the Questions section (pp. 3–...) first, lecture by lecture, without looking ahead.
2. Grade yourself against the Answer Key & Explanations section, which mirrors the question order.
3. Read the **Common trap** line for every question you miss — in this exam bank, the wrong options are deliberately built around the same mix-ups that show up on the real SA2, so the trap note is often more useful than the explanation itself.
4. Questions marked † are original (not verified past-paper) — treat them as high-confidence practice grounded in lecture content, not as guaranteed past-exam phrasing.

Question Distribution by Lecture Hours

Lecture	Hours	Questions	% of Exam
Lecture 17 — Motor System	1.0 hr	5	6.5%
Lecture 18 — Cerebellum	1.0 hr	5	6.5%
Lecture 19 — Cerebral Cortex	1.5 hr	8	10.4%
Lecture 20 — Basal Ganglia	1.0 hr	5	6.5%
Lecture 21 — Hypothalamus & Epithalamus	0.5 hr	3	3.9%
Lecture 22 — Thalamus	1.0 hr	5	6.5%
Lecture 23 — Limbic System & Olfaction	0.5 hr	3	3.9%
Lecture 24 — Autonomic Nervous System I (Structures)	1.0 hr	5	6.5%
Lecture 25 — Autonomic Nervous System II (Functions)	1.0 hr	5	6.5%
Lecture 26 — EEG & Sleep Physiology	1.0 hr	5	6.5%
Lecture 27 — Reflex Mechanism	1.0 hr	5	6.5%
Lecture 28 — Blood Supply of Brain & Meninges	1.5 hr	8	10.4%
Lecture 29 — Clinical Correlations	3.0 hr	15	19.5%
Total	15.0 hr	77	100%

Lecture 29 again carries the largest single share (15 questions, 19.5%), matching the 3.0 of 15 total hours the official schedule allocates to it. In this second set, its content spans visual-field and cranial-nerve localization vignettes, myasthenia gravis / basal ganglia / pain-pathway pharmacology, and original questions on spinal cord syndromes (Brown-Séquard, central cord), peripheral neuropathy patterns, and stroke-severity scoring drawn from the EP2 Neuro Exam lecture — deliberately different territory from Mock Exam 1's Lecture 29 content.

Section 1 — Questions (77 items, single best answer)

Items marked † are original questions written for this mock exam, grounded in lecture content but not verified past-paper items.

Lecture 17 — Motor System — 1.0 hr — 5 questions

1. When dynamic γ -motor neurons are co-activated with α -motor neurons during a voluntary contraction, the net effect on the muscle spindle's Ia afferent output is:

- A. Ia inhibition occurs
- B. Clonus occurs
- C. The muscle won't contract
- D. Fewer Ia impulses than with α -activation alone
- E. More Ia impulses than with α -activation alone

2. In the biceps reflex, the antagonist muscle is the:

- A. Afferent = Ib
- B. Antagonist = triceps
- C. Homonymous = hamstring
- D. Heteronymous = biceps
- E. Receptor = Golgi tendon organ

3. While walking on the beach, a patient suddenly steps on a sharp shell. Which sensory neuron/afferent fiber activates the flexor (withdrawal) reflex in the lower limb?

- A. A-delta
- B. Alpha motor neuron
- C. C fiber
- D. A-beta fiber
- E. Ia afferent fiber

4. The reflexive head-turning toward a sudden sound is mediated predominantly by the:

- A. Vestibulospinal tract
- B. Reticulospinal tract
- C. Tectospinal tract
- D. Spinothalamic tract
- E. Corticospinal tract

5. A lower motor neuron (LMN) lesion of the facial nerve results in:

- A. Contralateral lower face paralysis
- B. Ipsilateral lower face paralysis only
- C. Entire ipsilateral facial paralysis
- D. Contralateral upper face paralysis
- E. Bilateral lower face weakness

Lecture 18 — Cerebellum — 1.0 hr — 5 questions

6. A 60-year-old woman uses melatonin for insomnia. Which organ synthesizes and secretes melatonin?

- A. Liver
- B. Kidney
- C. Cerebellum
- D. Sweat gland
- E. Pineal gland

7. The inability to perform rapid alternating movements combined with an intention tremor is known as:

- A. Dysdiadochokinesia + intention tremor
- B. Ataxia alone
- C. Hemiballismus
- D. Resting tremor
- E. Chorea

8. Which cerebellar deep nucleus is primarily associated with motor planning?

- A. Fastigial
- B. Globose
- C. Emboliform
- D. Lateral dentate
- E. Medial dentate

9. Which of the following tracts can inhibit the cerebellar deep nuclei at a characteristically low, regular firing rate, distinctly slower than typical mossy fiber input?

- A. Spinocerebellar tract
- B. Olivocerebellar tract
- C. Tectocerebellar tract
- D. Vestibulocerebellar tract
- E. Corticopontocerebellar tract

10. The stabilizing circuit (vestibulocerebellum) synapses with which nucleus in the cerebellum?

- A. Dentate nucleus
- B. Emboliform nucleus
- C. Globose nucleus
- D. Fastigial nucleus
- E. Pontine nucleus

Lecture 19 — Cerebral Cortex — 1.5 hr — 8 questions

11. Which cortical cell layer provides the axons that form the corpus callosum?

- A. External granular
- B. External pyramidal (pyramidal cells)
- C. Internal granular
- D. Internal pyramidal
- E. Multiform

12. Olfactory information reaches primary olfactory (piriform) cortex in the temporal lobe via the:

- A. Stria terminalis
- B. Lateral olfactory stria
- C. Orbital cortex
- D. Anterior thalamic nucleus
- E. Septal nuclei

13. Bilateral destruction of the amygdala causes:

- A. Klüver-Bucy syndrome
- B. Korsakoff's syndrome
- C. Turner syndrome
- D. Temporal lobe epilepsy
- E. Horner's syndrome

14. A 71-year-old man presents with contralateral hemineglect, anosognosia, and visuospatial disability, but no aphasia. Which lobe is damaged?

- A. Frontal
- B. Parietal
- C. Occipital
- D. Temporal
- E. Limbic

15. The habenulointerpeduncular tract projects to the:

- A. Midbrain
- B. Thalamus
- C. Limbic lobe
- D. Insular cortex
- E. Neurohypophysis

16. A patient has abnormal arm and leg movements but remains fully conscious throughout the episode. This is a:

- A. Simple partial seizure
- B. Complex partial seizure
- C. Absence seizure
- D. Temporal lobe epilepsy
- E. Generalized tonic-clonic seizure

17. A patient has abnormal movement in their right arm and loses consciousness during the episode. What is the best description of the patient's condition?

- A. Simple partial seizure
- B. Complex partial seizure
- C. Tonic-clonic seizure
- D. Absence seizure
- E. Temporal lobe epilepsy

18. A patient understands language normally but cannot speak fluently, producing slow, effortful, telegraphic speech. The lesion is at:

- A. Wernicke's area
- B. Angular gyrus
- C. Broca's area
- D. Arcuate fasciculus
- E. Precentral gyrus

Lecture 20 — Basal Ganglia — 1.0 hr — 5 questions

19. The globus pallidus projects to the thalamus via:

- A. Ansa lenticularis
- B. Ansa peduncularis
- C. Fasciculus retroflexus
- D. Stria medullaris
- E. Stria terminalis

20. A woman suddenly collapses with tonic stiffening for about 30 seconds, followed by rhythmic clonic jerking. The diagnosis is:

- A. Complex partial seizure
- B. Simple partial seizure
- C. Tonic-clonic seizure
- D. Absence seizure
- E. Parkinson's disease

21. A 65-year-old man has a resting tremor, bradykinesia, shuffling gait, and decreased dopamine uptake on a PET scan. The diagnosis is:

- A. Hemiballismus
- B. Tardive dyskinesia
- C. Sydenham's chorea
- D. Parkinson's disease
- E. Huntington's disease

22. What is the most common cause of hemiballismus–hemichorea?

- A. Stroke
- B. Seizure
- C. Brain tumor
- D. Multiple sclerosis
- E. Traumatic brain injury

23. Which of the following indicates Parkinson's disease more specifically than other movement disorders?

- A. Unilateral resting tremor
- B. MRI findings
- C. Chorea
- D. Intention tremor
- E. Bilateral onset of symptoms

Lecture 21 — Hypothalamus & Epithalamus — 0.5 hr — 3 questions

24. Which hypothalamic nucleus lies in the tuberal region of the hypothalamus?

- A. Arcuate
- B. Supraoptic
- C. Mammillary
- D. Paraventricular
- E. Medial preoptic

25. Which statement about the functions of hypothalamic nuclei and their tracts is correct?

- A. Suprachiasmatic nuclei can send signals back to the retina via the optic tract.
- B. The mammillothalamic tract connects the dorsomedial nuclei and the mammillary body.
- C. The posterior hypothalamic nucleus sends sympathetic signals to the spinal cord (T1–L2) via the hypothalamospinal tract.
- D. The anterior hypothalamic nucleus cannot send parasympathetic signals to the Edinger-Westphal nucleus.
- E. The preoptic area is involved in thirst.

26. What increases the synthesis of melatonin?

- A. Inhibited suprachiasmatic nucleus
- B. Inhibited paraventricular nucleus
- C. Stimulation of optic tract
- D. Inhibited superior cervical ganglion
- E. Its control by parasympathetic system

Lecture 22 — Thalamus — 1.0 hr — 5 questions

27. A pain signal travels up the contralateral lateral spinothalamic tract toward the cortex. At which thalamic nucleus does this pathway synapse before its final relay to the primary somatosensory cortex?

- A. Centromedian nucleus
- B. Mediodorsal nucleus
- C. Ventral anterior nucleus
- D. Ventral lateral nucleus
- E. Ventral posterolateral nucleus

28. Basal ganglia and cerebellar output both reach the motor cortex only after first relaying through the thalamus. Which pair of thalamic nuclei serves this motor-relay role?

- A. Dorsomedial & lateral dorsal
- B. Ventral anterior & ventral lateral
- C. Lateral posterior & ventral posteromedial
- D. Intralaminar & midline
- E. Lateral geniculate & reticular

29. The gustatory (taste) pathway includes all of the following EXCEPT:

- A. Solitary nucleus (rostral part)
- B. VPM nucleus of the thalamus
- C. Geniculate/retrosal/nodose ganglia
- D. Insula and frontal operculum (gustatory cortex)
- E. Trapezoid body

30. Which thalamic nucleus is unique in that it does NOT project to the cerebral cortex, but instead forms a thin shell around the rest of the thalamus and modulates thalamocortical relay activity? †

- A. Pulvinar
- B. Reticular nucleus
- C. Centromedian nucleus
- D. Lateral geniculate nucleus
- E. Anterior nucleus

31. The largest nucleus of the thalamus, an associative (non-specific) nucleus important for visual attention and multimodal sensory integration, with extensive connections to parietal, temporal, and occipital association cortices, is the: †

- A. Ventral posterolateral nucleus
- B. Medial geniculate nucleus
- C. Pulvinar
- D. Centromedian nucleus
- E. Reticular nucleus

Lecture 23 — Limbic System & Olfaction — 0.5 hr — 3 questions

32. What is the relationship of the hippocampus to the inferior horn of the lateral ventricle?

- A. In floor
- B. In roof
- C. Rostral to ventricle
- D. In front
- E. No relation

33. Sara feels sudden fear and has an increased heart rate after hearing an unexpected sound at night. Which brain structure is most active in generating this response?

- A. Thalamus
- B. Basal ganglia
- C. Hypothalamus
- D. Amygdala
- E. Hippocampus

34. Which of the following statements about the fornix is correct?

- A. The precommissural fornix connects to the anterior nucleus of the thalamus.
- B. The postcommissural fornix connects to the mammillary body.
- C. The fornix originates from the amygdala.
- D. The fornix terminates in the cingulate gyrus.
- E. The fornix connects directly to the amygdala without passing through any commissure.

Lecture 24 — Autonomic Nervous System I (Structures) — 1.0 hr — 5 questions

35. Which of the following is NOT a feature of Horner's syndrome?

- A. Ptosis
- B. Anhidrosis
- C. Miosis
- D. Enophthalmos
- E. No saliva secretion

36. In the autonomic nervous system, preganglionic fibers release a neurotransmitter that binds to:

- A. N2 receptors
- B. M3 receptors
- C. α 1 receptors
- D. α 2 receptors
- E. β 2 receptors

37. Which of the following does NOT form the cardiac plexus?

- A. Superior cardiac branch
- B. Middle cardiac branch
- C. Inferior cardiac branch
- D. Postganglionic fibers T1–T4
- E. Greater splanchnic nerve

38. Which pairing of preganglionic neurons and their associated fibers/nerves is INCORRECT?

- A. Edinger–Westphal → Oculomotor nerve
- B. Superior salivatory → Greater petrosal nerve
- C. Inferior salivatory → Lesser petrosal nerve
- D. Dorsal motor nucleus of vagus → CN X
- E. T1–T4 → Superior cardiac branch

39. Which of the following is a parasympathetic fiber?

- A. Least splanchnic nerve
- B. Lumbar splanchnic nerve
- C. Sacral splanchnic nerve
- D. Pelvic splanchnic nerve
- E. Greater splanchnic nerve

Lecture 25 — Autonomic Nervous System II (Functions) — 1.0 hr — 5 questions

40. A patient has ingested a muscarinic (cholinergic) toxin from wild mushroom poisoning. Which combination of findings is expected?

- A. Increased salivation, bradycardia, relaxed urinary sphincter
- B. Decreased tears, tachycardia, contracted urinary sphincter
- C. Increased tears, bradycardia, relaxed urinary sphincter
- D. Increased salivation, tachycardia, contracted urinary sphincter
- E. Increased tears, bradycardia, contracted urinary sphincter

41. The enzyme that converts norepinephrine (NE) to epinephrine (Epi) in the adrenal medulla is:

- A. COMT
- B. MAO
- C. PNMT
- D. Tyrosine hydroxylase
- E. Dopamine β -hydroxylase

42. The baroreceptor reflex, which buffers acute changes in blood pressure, relies on afferent signals carried mainly by which two cranial nerves? †

- A. CN V and CN VII
- B. CN IX and CN X
- C. CN VII and CN IX
- D. CN X and CN XII
- E. CN III and CN VI

43. A patient with cardiac ischemia reports pain radiating to the left arm and jaw, even though the heart itself is the source of injury. This phenomenon — visceral pain perceived as arising from a somatic dermatome — is best explained by: †

- A. Gate-control theory
- B. Convergence-projection theory of referred pain
- C. Wind-up phenomenon
- D. Central sensitization
- E. Allodynia

44. Regarding autonomic control of salivary gland secretion, which statement is correct? †

- A. Only the parasympathetic system stimulates salivation; sympathetic input has no effect.
- B. Parasympathetic stimulation produces profuse, watery secretion; sympathetic stimulation produces a small-volume, thick/mucous secretion.
- C. Sympathetic stimulation produces profuse watery secretion; parasympathetic produces thick mucous secretion.
- D. Both divisions produce identical secretion, differing only in total volume.
- E. Salivary glands receive parasympathetic but not sympathetic innervation.

Lecture 26 — EEG & Sleep Physiology — 1.0 hr — 5 questions

45. Which brainwave pattern occurs during deep sleep (Stage N3)?

- A. Gamma
- B. Beta
- C. Alpha
- D. Theta
- E. Delta

46. The brain waves seen in a normal, awake, alert state with the eyes open are predominantly:

- A. Delta
- B. Beta
- C. Alpha
- D. Gamma
- E. Theta

47. Which EEG rhythm (4–7 Hz) is characteristically seen during drowsiness and light (Stage 1) sleep in adults, and is also a normal, prominent background rhythm during wakefulness in young children? †

- A. Delta
- B. Theta
- C. Alpha
- D. Beta
- E. Gamma

48. Sleep spindles and K-complexes, the EEG hallmarks used to score Stage 2 NREM sleep, arise primarily from oscillatory activity in the: †

- A. Suprachiasmatic nucleus
- B. Locus coeruleus
- C. Thalamus (thalamocortical circuitry)
- D. Hippocampus
- E. Raphe nuclei

49. During REM sleep, near-total skeletal muscle paralysis (atonia) is actively produced mainly through inhibition of spinal motor neurons originating from the: †

- A. Locus coeruleus
- B. Pontine/subcoeruleus region (via glycine/GABA-mediated inhibition)
- C. Suprachiasmatic nucleus
- D. Raphe nuclei (serotonergic)
- E. Red nucleus

Lecture 27 — Reflex Mechanism — 1.0 hr — 5 questions

50. The sensory receptor for the corneal reflex is the:

- A. Merkel disks
- B. Ruffini endings
- C. Free nerve endings
- D. Pacinian corpuscles
- E. Meissner corpuscles

51. What is the maximum possible score on the Glasgow Coma Scale (GCS)?

- A. 10
- B. 12
- C. 15
- D. 18
- E. 20

52. On the standard 0–4 scale for grading deep tendon reflexes, a reflex that is present but diminished compared to normal is graded: †

- A. Grade 0
- B. Grade 1
- C. Grade 2
- D. Grade 3
- E. Grade 4

53. A patient shows absent superficial abdominal reflexes together with exaggerated (brisk) deep tendon reflexes in the lower limbs. This dissociation is most suggestive of a lesion in the: †

- A. Peripheral nerve
- B. Corticospinal tract
- C. Cerebellum
- D. Neuromuscular junction
- E. Muscle

54. Flicking the distal phalanx of the middle finger and observing reflexive flexion of the thumb and index finger is the Hoffmann sign, and its presence (particularly if asymmetric or accompanied by other findings) is considered: †

- A. A normal finding in all healthy adults
- B. A pathological reflex suggesting a cervical corticospinal (UMN) lesion
- C. Diagnostic of peripheral neuropathy
- D. A sign of cerebellar dysfunction
- E. Equivalent in significance to a normal ankle reflex

Lecture 28 — Blood Supply of Brain & Meninges — 1.5 hr — 8 questions

55. Cerebrospinal fluid from the ventricular system first communicates with the subarachnoid space via which structure?

- A. 4th ventricle
- B. 3rd ventricle
- C. Subarachnoid granulations
- D. Choroid plexus
- E. Cerebral aqueduct

56. A 55-year-old male, post-stroke, is later found on autopsy to have a shaded lesion in the lower pons. The embolus responsible is most likely in which artery?

- A. ACA
- B. MCA
- C. PICA
- D. AICA
- E. Basilar artery

57. Which of the following statements about stroke is TRUE?

- A. TIA often lasts 2–3 weeks
- B. Ischemic stroke results from rupture of brain blood vessels
- C. Hemorrhagic stroke is due to plaques/fat lining the vessel wall
- D. AVMs (arteriovenous malformations) are local cerebral clots
- E. Stroke is neurological injury from occlusion or rupture of cerebral vessels

58. Regarding cerebral cortex blood supply, which statement is true?

- A. MCA: contralateral arm, leg, speech areas
- B. ACA: contralateral leg and micturition
- C. MCA: ipsilateral arm, face, vision
- D. PCA: ipsilateral vision
- E. ACA: contralateral leg, auditory, speech

59. Which of the following statements about stroke is FALSE?

- A. Stroke = neurological injury from occlusion or rupture of cerebral vessels
- B. A TIA (transient ischemic attack) often lasts 2–3 weeks
- C. Hemorrhagic stroke results from rupture of a cerebral vessel
- D. Ischemic stroke results from occlusion of a cerebral vessel
- E. AVMs (arteriovenous malformations) are a structural vascular abnormality, not a clot

60. A 71-year-old man has a large left MCA territory stroke. An embolus from which vessel is the most likely underlying source?

- A. ACA
- B. MCA
- C. PCA
- D. ICA
- E. Vertebral artery

61. Which statement about intracranial venous drainage is correct?

- A. Superior sagittal sinus → great cerebral vein
- B. Superior cerebral veins → superior sagittal sinus
- C. Inferior sagittal sinus → cavernous sinus
- D. Great cerebral vein → superior sagittal sinus
- E. Cavernous sinus → superior sagittal sinus directly

62. A patient has an embolus originating from the internal carotid artery. In which branch is the embolus most likely to lodge?

- A. MCA
- B. PCOM
- C. ACOM
- D. ACA
- E. VA

Lecture 29 — Clinical Correlations — 3.0 hr — 15 questions

63. A 40-year-old man has presbyopia (difficulty reading fine print up close). Which statement about the near-focusing (accommodation) reflex is correct?

- A. The lens becomes flatter during near focus.
- B. Accommodation is controlled by the sympathetic nervous system.
- C. M3 receptors mediate contraction of the ciliary muscle.
- D. α_2 receptors contract the pupillary sphincter muscle.
- E. Ciliary muscle contraction increases suspensory ligament tension.

64. A patient with chorea, basal ganglia atrophy on imaging, and a family history of similar symptoms is suspected of having Huntington's disease. What is the best diagnostic test to confirm this?

- A. CT scan
- B. MRI
- C. Genetic testing
- D. Blood glucose
- E. EMG (electromyography)

65. Which sympathetic receptor is responsible for contracting the internal urinary sphincter, supporting continence during bladder filling?

- A. Alpha-1
- B. Beta-3
- C. Beta-2
- D. NM
- E. M3

66. What type of fiber provides the major excitatory input to the neostriatum (caudate and putamen)?

- A. Nigrostriatal, inhibitory via GABA
- B. Corticostriatal, excitatory via Glutamate
- C. Nigrostriatal, excitatory via Glutamate
- D. Thalamostriatal, inhibitory via GABA
- E. Pallidostriatal, inhibitory via GABA

67. The ganglia housing the first-order (peripheral) taste neurons are found in which of the following structures?

- A. Geniculate, Petrosal, Nodose ganglia
- B. Superior, Ciliary, Otic ganglia
- C. Stellate, Geniculate, Nodose ganglia
- D. Trigeminal, Geniculate, Nodose ganglia
- E. Geniculate, Spiral, Pterygopalatine ganglia

68. In the Gate-control theory of pain, opioid peptides such as enkephalins reduce nociceptive transmission primarily by:

- A. Blocking Na⁺ channels presynaptically
- B. Opening presynaptic Ca²⁺ channels
- C. Inhibiting Ca²⁺ influx and opening K⁺ channels
- D. Stimulating NMDA receptors
- E. Enhancing substance P release

69. A patient with myasthenia gravis is being managed medically. Which of the following is NOT an appropriate treatment for myasthenia gravis?

- A. Anticholinesterase drugs
- B. Prednisone
- C. Plasmapheresis
- D. Thymectomy
- E. Dopamine agonists

70. A lesion of the left Meyer's loop (the temporal lobe portion of the optic radiation) results in:

- A. Left homonymous hemianopia
- B. Right superior quadrantanopia
- C. Right inferior quadrantanopia
- D. Bitemporal hemianopia
- E. Left superior quadrantanopia

71. Which cranial nerve does NOT contribute afferent fibers to the spinal trigeminal nucleus?

- A. CN V
- B. CN VII
- C. CN IX
- D. CN X
- E. CN XI

72. A lesion of the left dorsal (parietal) optic radiation produces:

- A. Right superior quadrantanopia
- B. Right inferior quadrantanopia
- C. Right homonymous hemianopia
- D. Bitemporal hemianopia
- E. Central scotoma

73. A patient sustains a stab wound causing hemisection of the spinal cord (Brown-Séquard syndrome). On the side of the injury, motor function and vibration/proprioception are lost; on the opposite side, pain and temperature sensation are lost below the level of injury. This lateralization pattern occurs because: †

- A. The corticospinal and dorsal column tracts have not yet crossed at the level of injury, while the spinothalamic tract already crossed near its segmental entry.
- B. All three tracts cross at the level of injury.
- C. The spinothalamic tract does not cross at any level.
- D. The dorsal columns cross in the spinal cord, unlike the corticospinal tract.
- E. Brown-Séquard syndrome reflects a lesion of peripheral nerves, not the spinal cord.

74. An elderly man falls and sustains a hyperextension neck injury. He develops weakness that is much greater in the arms than the legs, with a 'cape-like' loss of pain and temperature sensation across the shoulders and upper arms, while leg strength and lower-body sensation are relatively preserved. This is most consistent with: †

- A. Brown-Séquard syndrome
- B. Anterior cord syndrome
- C. Central cord syndrome
- D. Subacute combined degeneration
- E. Guillain-Barré syndrome

75. A patient with a chronic, length-dependent polyneuropathy shows sensory loss and weakness that is noticeably worse distally (feet) than proximally. This distal-predominant pattern is most typical of which underlying process? †

- A. Demyelinating polyneuropathy
- B. Axonal polyneuropathy
- C. Mononeuropathy
- D. NMJ disorder
- E. Myopathy

76. When testing vibration sense with a 128 Hz tuning fork, which of the following describes the correct technique? †

- A. Ask the patient 'Do you feel that?' immediately after striking the fork.
- B. Place the struck, vibrating fork on a bony prominence (e.g., the great toe) and ask the patient to report when the vibration stops.
- C. Place the non-vibrating fork on the skin as a control before testing.
- D. Test only proximal joints, since distal vibration sense is normally unreliable.
- E. Use a 512 Hz fork for lower-limb testing to increase sensitivity.

77. The NIH Stroke Scale (NIHSS), used at the bedside in acute stroke care, is used primarily to: †

- A. Diagnose the specific vascular territory of a stroke
- B. Quantify stroke severity and help guide/monitor treatment such as thrombolysis
- C. Replace the need for brain imaging before treatment
- D. Predict long-term cognitive outcome only
- E. Determine eligibility for surgical clipping of aneurysms

Section 2 — Answer Key & Explanations

Every entry states the correct answer, why it's correct, the specific wrong-answer trap built into the distractors, and where the question came from.

Lecture 17 — Motor System — 1.0 hr — 5 questions

Q1. When dynamic γ -motor neurons are co-activated with α -motor neurons during a voluntary contraction, the net effect on the muscle spindle's Ia afferent output is:

Correct answer: E. More Ia impulses than with α -activation alone

Alpha-gamma coactivation keeps the muscle spindle taut even as the whole muscle shortens (extrafusal fibers contract under alpha drive while intrafusal fibers are simultaneously “taken up the slack” by gamma drive), so the spindle continues firing — in fact generating MORE Ia impulses than alpha activation alone would produce.

Common trap: “Ia inhibition” (A) and “clonus” (B) both sound like plausible “abnormal circuit” answers, but alpha-gamma coactivation is a normal physiological mechanism, not a pathological one — don't let unfamiliar terminology pull you toward pathology-flavored distractors.

Verified — corrected past-paper key (Q2)

Q2. In the biceps reflex, the antagonist muscle is the:

Correct answer: B. Antagonist = triceps

For the biceps reflex, striking the biceps tendon stretches the biceps (Ia afferent), which excites the biceps itself (homonymous) while inhibiting its antagonist, the triceps.

Common trap: Options C and D misuse “homonymous” and “heteronymous” by attaching them to the wrong muscles (hamstring has nothing to do with an arm reflex) — read each option literally rather than pattern-matching to the knee-jerk reflex's terminology.

Verified — corrected past-paper key (Q8)

Q3. While walking on the beach, a patient suddenly steps on a sharp shell. Which sensory neuron/afferent fiber activates the flexor (withdrawal) reflex in the lower limb?

Correct answer: A. A-delta

A-delta fibers carry the fast, sharp, well-localized “first pain” signal that triggers the rapid flexor (withdrawal) reflex, pulling the limb away before the slower, duller C-fiber pain signal even arrives.

Common trap: C fibers (C) do carry pain too, but they are slow and carry the delayed, burning “second pain” — too slow to be the fiber responsible for the fast, protective withdrawal reflex itself.

Verified — corrected past-paper key (Q13)

Q4. The reflexive head-turning toward a sudden sound is mediated predominantly by the:

Correct answer: C. Tectospinal tract

The reflexive head/eye turn toward a sudden sound is mediated by the tectospinal tract (inferior colliculus → tectospinal tract → cervical spinal motor nuclei), a fast, subcortical orienting reflex pathway.

Common trap: The vestibulospinal tract (A) instead handles postural/antigravity reflexes triggered by head position and movement, not orienting toward external sounds — a common mix-up between two brainstem “reflex” tracts.

Verified — corrected past-paper key (Q26)

Q5. A lower motor neuron (LMN) lesion of the facial nerve results in:

Correct answer: C. Entire ipsilateral facial paralysis

An LMN facial nerve lesion (e.g., Bell's palsy) damages the entire facial nerve after it has already left the pons, so it causes COMPLETE ipsilateral facial paralysis — both upper and lower face — unlike a UMN lesion, which spares the upper face due to bilateral cortical innervation of the forehead.

Common trap: “Ipsilateral lower face paralysis only” (B) describes the UMN pattern, not the LMN one — an easy way to accidentally answer the opposite question.

Verified — corrected past-paper key (Q27)

Lecture 18 — Cerebellum — 1.0 hr — 5 questions

Q6. A 60-year-old woman uses melatonin for insomnia. Which organ synthesizes and secretes melatonin?

Correct answer: E. Pineal gland

The pineal gland produces and secretes melatonin, using signals relayed from the suprachiasmatic nucleus via a sympathetic pathway through the superior cervical ganglion.

Common trap: “Cerebellum” (C) has no established role in melatonin synthesis or secretion — a distractor that simply doesn't belong in this physiological pathway at all.

Verified — corrected past-paper key (Q30)

Q7. The inability to perform rapid alternating movements combined with an intention tremor is known as:

Correct answer: A. Dysdiadochokinesia + intention tremor

Dysdiadochokinesia (inability to perform rapid alternating movements) plus intention tremor are both classic cerebellar hemisphere signs and are commonly tested together as a pair.

Common trap: “Ataxia alone” (B) is too generic — the question specifically names two distinct findings, which together have their own specific combined name rather than the broad umbrella term “ataxia.”

Verified — corrected past-paper key (Q35)

Q8. Which cerebellar deep nucleus is primarily associated with motor planning?

Correct answer: D. Lateral dentate

The lateral cerebellar hemisphere and its associated lateral dentate nucleus (via the cerebrocerebellum) are involved in motor planning, in contrast to the medial vermis/fastigial nucleus (posture/balance) or the intermediate zone/interposed nuclei (ongoing limb execution).

Common trap: Fastigial (A) is a tempting “famous nucleus” answer, but it belongs to the medial vestibulocerebellar/postural system, functionally opposite to the planning-related lateral hemisphere.

Verified — corrected past-paper key (Q38)

Q9. Which of the following tracts can inhibit the cerebellar deep nuclei at a characteristically low, regular firing rate, distinctly slower than typical mossy fiber input?

Correct answer: B. Olivocerebellar tract

Climbing fibers of the olivocerebellar tract are well known for their characteristically low, regular firing rate (commonly cited in the range of roughly 1–10 Hz, depending on the source) — a distinguishing feature versus the much higher and more variable firing rates of mossy fiber input from other pathways.

Common trap: Spinocerebellar and corticopontocerebellar pathways (A, E) both use mossy fibers, which fire at much higher and more variable rates — not the characteristic low, regular firing rate being asked about here. (Exact quoted values for climbing fiber rate vary somewhat by textbook; the qualitative distinction from mossy fiber input is the reliable, testable fact.)

Verified — corrected past-paper key (Q40)

Q10. The stabilizing circuit (vestibulocerebellum) synapses with which nucleus in the cerebellum?

Correct answer: D. Fastigial nucleus

The fastigial nucleus is the deep cerebellar nucleus associated with the vestibulocerebellum (flocculonodular lobe) / stabilizing circuit, receiving Purkinje cell output from the vestibulocerebellar cortex and projecting onward to the vestibular nuclei.

Common trap: Dentate (A) is the largest and most “famous” deep nucleus, tempting students to pick it reflexively, but it belongs to the lateral hemisphere / cerebrocerebellar (planning) circuit, not the medial stabilizing circuit.

Verified — corrected past-paper key (Q43)

Lecture 19 — Cerebral Cortex — 1.5 hr — 8 questions

Q11. Which cortical cell layer provides the axons that form the corpus callosum?

Correct answer: B. External pyramidal (pyramidal cells)

Commissural (corpus callosum) axons classically arise from layer III (external pyramidal) neurons, which connect homologous areas of the two hemispheres.

Common trap: Internal pyramidal (layer V, option D) neurons instead give rise to long PROJECTION fibers (e.g., the corticospinal tract descending to the spinal cord) — a different fiber category entirely from commissural connections.

Verified — corrected past-paper key (Q52)

Q12. Olfactory information reaches primary olfactory (piriform) cortex in the temporal lobe via the:

Correct answer: B. Lateral olfactory stria

Olfactory information reaches primary (piriform) cortex via the lateral olfactory stria — uniquely bypassing the thalamus, unlike every other sensory modality.

Common trap: The stria terminalis (A) is the major output/input pathway of the amygdala, not the primary route by which raw olfactory signals reach cortex — an easy “stria” name mix-up.

Verified — corrected past-paper key (Q55)

Q13. Bilateral destruction of the amygdala causes:

Correct answer: A. Klüver-Bucy syndrome

Bilateral amygdala destruction (part of the bilateral temporal lobe damage seen in Klüver-Bucy syndrome) produces the syndrome's characteristic hyperorality, hypersexuality, placidity, and psychic blindness.

Common trap: *Korsakoff's syndrome (B) is a memory disorder tied to mammillary body/Papez circuit damage (classically from thiamine deficiency), an entirely different structure and mechanism from amygdala destruction.*

Verified — corrected past-paper key (Q62)

Q14. A 71-year-old man presents with contralateral hemineglect, anosognosia, and visuospatial disability, but no aphasia. Which lobe is damaged?

Correct answer: B. Parietal

Hemineglect, anosognosia (unawareness of one's own deficit), and visuospatial disability without aphasia is the classic (typically non-dominant, right) parietal lobe syndrome.

Common trap: *Because language is spared, it's tempting to rule out the dominant hemisphere entirely and guess a "language-adjacent" lobe like frontal or temporal — but the specific triad described here is a parietal, not frontal or temporal, signature.*

Verified — corrected past-paper key (Q65)

Q15. The habenulointerpeduncular tract projects to the:

Correct answer: A. Midbrain

The habenulointerpeduncular tract (fasciculus retroflexus) connects the habenula to the interpeduncular nucleus — a midbrain structure, as its own name suggests.

Common trap: *"Neurohypophysis" (E) is reached instead via a completely different pathway (the supraopticohypophyseal tract), which has nothing to do with the habenula.*

Verified — corrected past-paper key (Q66)

Q16. A patient has abnormal arm and leg movements but remains fully conscious throughout the episode. This is a:

Correct answer: A. Simple partial seizure

By definition, abnormal movement WITH preserved consciousness is a simple partial (focal aware) seizure; complex partial seizures, by contrast, specifically require impaired consciousness.

Common trap: *"Temporal lobe epilepsy" (D) names a location, not a consciousness-based classification, and temporal lobe seizures more often impair consciousness (making them typically complex, not simple, partial) — so it doesn't correctly answer what's being asked here.*

Verified — corrected past-paper key (Q67)

Q17. A patient has abnormal movement in their right arm and loses consciousness during the episode. What is the best description of the patient's condition?

Correct answer: B. Complex partial seizure

If consciousness IS lost during a focal-onset seizure with limb movement, this is by definition a complex partial seizure — the presence of impaired awareness is exactly what distinguishes it from a simple partial seizure.

Common trap: *"Tonic-clonic seizure" (C) is tempting because it's the most "famous" seizure type, but that term specifically implies a generalized, bilateral stiffening-then-jerking pattern, not a focal seizure with impaired awareness as described here.*

Verified — corrected past-paper key (Q75)

Q18. A patient understands language normally but cannot speak fluently, producing slow, effortful, telegraphic speech. The lesion is at:

Correct answer: C. Broca's area

Preserved comprehension with non-fluent, effortful speech is the classic definition of Broca's (motor, expressive) aphasia, localizing to Broca's area in the dominant inferior frontal gyrus.

Common trap: Arcuate fasciculus (D) damage instead produces conduction aphasia, which is fluent (not effortful) but has poor repetition — a different, easily confused aphasia subtype.

Verified — corrected past-paper key (Q79)

Lecture 20 — Basal Ganglia — 1.0 hr — 5 questions

Q19. The globus pallidus projects to the thalamus via:

Correct answer: A. Ansa lenticularis

The globus pallidus (internal segment) projects to the thalamus (VA/VL) via the ansa lenticularis, together with the lenticular fasciculus, ultimately joining the thalamic fasciculus.

Common trap: "Fasciculus retroflexus" (C) is the habenulointerpeduncular tract instead, an unrelated limbic pathway that simply sounds similarly "anatomical" and technical.

Verified — corrected past-paper key (Q81)

Q20. A woman suddenly collapses with tonic stiffening for about 30 seconds, followed by rhythmic clonic jerking. The diagnosis is:

Correct answer: C. Tonic-clonic seizure

Tonic stiffening followed by rhythmic clonic jerking is the classic generalized tonic-clonic (grand mal) seizure pattern, typically with loss of consciousness throughout.

Common trap: Absence seizure (D) is a tempting "generalized seizure" distractor, but it presents as a brief staring spell with no tonic or clonic movement at all — a completely different clinical picture from what's described here.

Verified — corrected past-paper key (Q84)

Q21. A 65-year-old man has a resting tremor, bradykinesia, shuffling gait, and decreased dopamine uptake on a PET scan. The diagnosis is:

Correct answer: D. Parkinson's disease

Resting tremor, bradykinesia, shuffling gait, and decreased dopaminergic PET signal together are the classic clinical and imaging picture of Parkinson's disease.

Common trap: Huntington's disease (E) is a hyperkinetic (chorea) disorder with basal ganglia atrophy, essentially the opposite clinical picture (excess movement, not bradykinesia) from what's described.

Verified — corrected past-paper key (Q86)

Q22. What is the most common cause of hemiballismus–hemichorea?

Correct answer: A. Stroke

Hemiballismus-hemichorea is most commonly caused by a small (often lacunar) stroke affecting the contralateral subthalamic nucleus.

Common trap: Brain tumor (C) can occasionally cause similar movement disorders by mass effect, but vascular lesions (stroke) of the STN are by far the most common overall cause, especially in the classic exam vignette.

Verified — corrected past-paper key (Q87)

Q23. Which of the following indicates Parkinson's disease more specifically than other movement disorders?

Correct answer: A. Unilateral resting tremor

Parkinson's disease classically begins unilaterally with a resting tremor; chorea points toward Huntington's disease, intention tremor toward cerebellar disease, and MRI is typically unremarkable in Parkinson's, which remains a clinical diagnosis.

Common trap: "Bilateral onset" (E) is nearly the opposite of the classic Parkinson's presentation, which is notably asymmetric/unilateral at onset — a detail that's easy to overlook.

Verified — corrected past-paper key (Q90)

Lecture 21 — Hypothalamus & Epithalamus — 0.5 hr — 3 questions

Q24. Which hypothalamic nucleus lies in the tuberal region of the hypothalamus?

Correct answer: A. Arcuate

The arcuate nucleus lies in the tuberal (middle) region of the hypothalamus, along with the ventromedial and dorsomedial nuclei, and is the modern integration center for hunger/appetite signaling (NPY/AgRP and POMC/CART neurons).

Common trap: Supraoptic and paraventricular nuclei (B, D) both lie in the anterior/supraoptic region, not the tuberal region — an easy region mix-up if you're relying on nucleus names alone rather than their actual anteroposterior location.

Verified — corrected past-paper key (Q94)

Q25. Which statement about the functions of hypothalamic nuclei and their tracts is correct?

Correct answer: C. The posterior hypothalamic nucleus sends sympathetic signals to the spinal cord (T1–L2) via the hypothalamospinal tract.

The posterior hypothalamic nucleus genuinely does send descending sympathetic signals to the spinal cord (T1–L2) via the hypothalamospinal tract — the anatomical basis for why a posterior hypothalamic/brainstem lesion along this pathway can cause Horner's syndrome.

Common trap: Option B sounds plausible because it correctly names the mammillothalamic tract, but misattributes its target — the tract actually connects the mammillary body to the ANTERIOR thalamic nucleus (the Papez circuit relay), not the dorsomedial nucleus.

Verified — corrected past-paper key (Q99)

Q26. What increases the synthesis of melatonin?

Correct answer: A. Inhibited suprachiasmatic nucleus

The melatonin pathway runs: darkness → reduced retinal input → reduced suprachiasmatic nucleus activity → disinhibition of the paraventricular nucleus → increased sympathetic output via the superior cervical ganglion → increased pineal melatonin synthesis. An inhibited (i.e., less active) SCN is therefore the initiating event that allows melatonin synthesis to rise.

Common trap: "Stimulation of optic tract" (C) implies light exposure, which should SUPPRESS, not increase, melatonin — a direction-of-effect reversal that's easy to fall for.

Verified — corrected past-paper key (Q100)

Lecture 22 — Thalamus — 1.0 hr — 5 questions

Q27. A pain signal travels up the contralateral lateral spinothalamic tract toward the cortex. At which thalamic nucleus does this pathway synapse before its final relay to the primary somatosensory cortex?

Correct answer: E. Ventral posterolateral nucleus

VPL (ventral posterolateral nucleus) is the classic thalamic relay for body pain, temperature, and touch, receiving the contralateral spinothalamic tract as well as the dorsal column-medial lemniscus pathway.

Common trap: *Mediodorsal nucleus (B) is a very common distractor here because it “sounds” somatosensory, but it actually relays limbic/prefrontal circuitry, not spinothalamic input.*

Verified — corrected past-paper key (Q107)

Q28. Basal ganglia and cerebellar output both reach the motor cortex only after first relaying through the thalamus. Which pair of thalamic nuclei serves this motor-relay role?

Correct answer: B. Ventral anterior & ventral lateral

VA (receiving basal ganglia output) and VL (receiving cerebellar/dentate output) are the thalamic motor relay nuclei, projecting onward to motor and premotor cortex.

Common trap: *“Lateral geniculate & reticular” (E) sound anatomically similar to VA/VL by virtue of both containing “lateral,” but the lateral geniculate nucleus is purely visual and has no motor cortex connection.*

Verified — corrected past-paper key (Q109)

Q29. The gustatory (taste) pathway includes all of the following EXCEPT:

Correct answer: E. Trapezoid body

The trapezoid body is part of the auditory pathway (carrying crossing fibers from the cochlear nuclei), not the gustatory pathway — the true taste pathway runs from the cranial nerve ganglia through the rostral solitary nucleus, VPM, and on to the gustatory cortex.

Common trap: *It's tempting to assume every brainstem-sounding relay structure in a sensory-pathway list belongs to that pathway, but the trapezoid body is specifically an auditory landmark that has no role in taste.*

Verified — corrected past-paper key (Q113)

Q30. Which thalamic nucleus is unique in that it does NOT project to the cerebral cortex, but instead forms a thin shell around the rest of the thalamus and modulates thalamocortical relay activity?

Correct answer: B. Reticular nucleus

The thalamic reticular nucleus is a thin, GABAergic shell that surrounds the rest of the thalamus, receiving collaterals from both thalamocortical and corticothalamic fibers and inhibiting relay neurons — it is the one thalamic nucleus with no direct projection to cortex, acting instead as a “gatekeeper” for attention and arousal-related gating of sensory relay.

Common trap: *Pulvinar (A) is tempting because it's also “different” from a typical specific relay nucleus, but the pulvinar in fact projects extensively TO cortex (parietal, temporal, occipital association areas) — the opposite property from the reticular nucleus.*

Original — standard thalamic neuroanatomy within the Lecture 22 (Thalamus) scope. †

Q31. The largest nucleus of the thalamus, an associative (non-specific) nucleus important for visual attention and multimodal sensory integration, with extensive connections to parietal, temporal, and occipital association cortices, is the:

Correct answer: C. Pulvinar

The pulvinar is the largest thalamic nucleus. As an associative nucleus, it does not simply relay one specific primary sensory modality but instead integrates multimodal information and supports visual attention, with broad projections to posterior association cortices.

Common trap: Medial geniculate (B) is tempting as a “large-sounding” name, but it is actually a small, dedicated primary auditory relay nucleus — the opposite of the large, non-specific, associative pulvinar.

Original — standard thalamic neuroanatomy within the Lecture 22 (Thalamus) scope. †

Lecture 23 — Limbic System & Olfaction — 0.5 hr — 3 questions

Q32. What is the relationship of the hippocampus to the inferior horn of the lateral ventricle?

Correct answer: A. In floor

The hippocampus forms a prominent bulge in the floor of the inferior (temporal) horn of the lateral ventricle — a key landmark for identifying it on axial/coronal imaging.

Common trap: “In roof” (B) reverses the correct anatomical relationship — the choroid plexus, not the hippocampus, occupies part of the roof of this ventricular segment.

Verified — corrected past-paper key (Q116)

Q33. Sara feels sudden fear and has an increased heart rate after hearing an unexpected sound at night. Which brain structure is most active in generating this response?

Correct answer: D. Amygdala

Fear and its associated autonomic response (increased heart rate) is the classic signature of amygdala activation, which rapidly links a sensory stimulus to an emotional/autonomic response, sometimes even before full conscious processing is complete.

Common trap: The hypothalamus (C) does execute the downstream autonomic/endocrine changes of fear, but the structure that most directly generates and coordinates the fear response itself is the amygdala, one step upstream.

Verified — corrected past-paper key (Q121)

Q34. Which of the following statements about the fornix is correct?

Correct answer: B. The postcommissural fornix connects to the mammillary body.

The postcommissural fornix carries hippocampal output onward to the mammillary body, completing an essential leg of the Papez memory circuit.

Common trap: The precommissural fornix (A) instead projects mainly to the septal area, not directly to the anterior thalamic nucleus — the anterior nucleus is reached later, via the mamillothalamic tract from the mammillary body, not directly from the fornix itself.

Verified — corrected past-paper key (Q125)

Lecture 24 — Autonomic Nervous System I (Structures) — 1.0 hr — 5 questions

Q35. Which of the following is NOT a feature of Horner's syndrome?

Correct answer: E. No saliva secretion

Horner's syndrome results from disruption of sympathetic innervation to the eye and face, producing ptosis, miosis, and anhidrosis (with a mild, apparent enophthalmos from the narrowed palpebral fissure). Salivation, however, is controlled predominantly by parasympathetic secretomotor pathways that are not part of the sympathetic oculo-facial chain — so “no saliva secretion” is not a feature of Horner's syndrome.

Common trap: *Enophthalmos (D) looks like the tempting “odd one out” since it's a subtler, less textbook-famous finding than ptosis/miosis/anhidrosis, but it is a genuine (if apparent) Horner's finding — the truly unrelated option is the one describing salivary function.*

Verified, resolved — corrected past-paper key (Q129). The original archive extraction left this item ambiguous; resolved here using the principle that salivation is a parasympathetic, not sympathetic, function.

Q36. In the autonomic nervous system, preganglionic fibers release a neurotransmitter that binds to:

Correct answer: A. N2 receptors

All preganglionic autonomic fibers, sympathetic and parasympathetic alike, release acetylcholine acting on nicotinic (N2) receptors at the ganglionic synapse.

Common trap: *M3 (B) is the classic postganglionic parasympathetic receptor at the effector organ — a tempting choice, but it is the wrong synapse for this question (preganglionic, not postganglionic).*

Verified — corrected past-paper key (Q132)

Q37. Which of the following does NOT form the cardiac plexus?

Correct answer: E. Greater splanchnic nerve

The greater splanchnic nerve supplies abdominal viscera via the celiac ganglion and has no role in the cardiac plexus, which is instead formed by the superior/middle/inferior cardiac branches and postganglionic sympathetic fibers from T1–T4.

Common trap: *All three “cardiac branch” options (A–C) sound like an obvious set to pick from, but the question asks which does NOT contribute — don't let the presence of “cardiac” in three option names distract from correctly identifying the abdominal outlier.*

Verified — corrected past-paper key (Q133)

Q38. Which pairing of preganglionic neurons and their associated fibers/nerves is INCORRECT?

Correct answer: E. T1–T4 → Superior cardiac branch

Preganglionic sympathetic fibers from T1–T4 first synapse in the cervical sympathetic ganglia; it is the POSTganglionic fibers that then form the actual cardiac branches. Directly pairing the preganglionic segmental level with the peripheral cardiac branch skips this synapse and is therefore the incorrect pairing.

Common trap: *Options A–D are all textbook-correct parasympathetic preganglionic-to-nerve pairings, which can make E feel like it “belongs” in the same style of list — but sympathetic anatomy requires an extra ganglionic relay that the other pairings don't.*

Verified — corrected past-paper key (Q135)

Q39. Which of the following is a parasympathetic fiber?

Correct answer: D. Pelvic splanchnic nerve

Of all the named splanchnic nerves, only the pelvic splanchnic nerve (S2–S4) is parasympathetic; the greater, lesser, least, lumbar, and sacral splanchnic nerves are all sympathetic.

Common trap: “Sacral splanchnic nerve” (C) sounds like it should be the parasympathetic one since it's named after the sacral (parasympathetic outflow) region, but the sacral splanchnic nerves are actually sympathetic chain branches — it's specifically the PELVIC splanchnic nerve that carries the parasympathetic fibers.

Verified — corrected past-paper key (Q137)

Lecture 25 — Autonomic Nervous System II (Functions) — 1.0 hr — 5 questions

Q40. A patient has ingested a muscarinic (cholinergic) toxin from wild mushroom poisoning. Which combination of findings is expected?

Correct answer: A. Increased salivation, bradycardia, relaxed urinary sphincter

Muscarinic (cholinergic) toxicity produces the classic SLUDGE picture — Salivation, Lacrimation, Urination, Defecation, GI upset, Emesis — alongside bradycardia; excess parasympathetic tone also relaxes the internal urinary sphincter, facilitating urination.

Common trap: Options C and E both correctly pair bradycardia with excess secretion, differing only in whether salivation or tears is named; in reality both increase, but “salivation” is the classically emphasized first letter of the SLUDGE mnemonic and the relaxed (not contracted) sphincter is what allows urination — make sure both details, not just one, match the muscarinic pattern.

Verified, resolved — corrected past-paper key (Q138). The original archive extraction left this item ambiguous between two similar options; resolved here using the classic SLUDGE muscarinic toxidrome.

Q41. The enzyme that converts norepinephrine (NE) to epinephrine (Epi) in the adrenal medulla is:

Correct answer: C. PNMT

PNMT (phenylethanolamine N-methyltransferase) converts norepinephrine to epinephrine — the final step of catecholamine synthesis, present specifically in the adrenal medulla due to local cortisol induction of the enzyme.

Common trap: Dopamine β -hydroxylase (E) is a tempting distractor because it's also a real catecholamine-synthesis enzyme, but it catalyzes the EARLIER step (dopamine → norepinephrine), not the final NE → epinephrine conversion being asked about here.

Verified — corrected past-paper key (Q141)

Q42. The baroreceptor reflex, which buffers acute changes in blood pressure, relies on afferent signals carried mainly by which two cranial nerves?

Correct answer: B. CN IX and CN X

Baroreceptors in the carotid sinus signal centrally via CN IX (glossopharyngeal, through Hering's nerve), and baroreceptors in the aortic arch signal via CN X (vagus); both afferents converge on the nucleus tractus solitarius to reflexively adjust heart rate and vascular tone.

Common trap: CN VII (facial, option A/C) is numerically “next to” IX and carries its own parasympathetic (salivary/lacrima) and taste fibers, which tempts students to lump it in with the baroreceptor pathway — but it plays no role in baroreceptor signaling.

Original — standard ANS physiology within the Lecture 25 (Functions of the ANS) scope. †

Q43. A patient with cardiac ischemia reports pain radiating to the left arm and jaw, even though the heart itself is the source of injury. This phenomenon — visceral pain perceived as arising from a somatic dermatome — is best explained by:

Correct answer: B. Convergence-projection theory of referred pain

Visceral and somatic afferents from the same spinal segment converge onto shared second-order (spinothalamic) neurons; because the brain is far more accustomed to interpreting somatic input from that segment, it misattributes the visceral signal to the corresponding somatic dermatome — the convergence-projection theory of referred pain.

Common trap: Gate-control theory (A) is a very commonly confused distractor here, but it explains how non-painful input can suppress pain transmission (the mechanism behind “rubbing an injury to ease the pain”) — a different mechanism entirely from why pain is mislocalized to begin with.

Original — standard ANS/pain physiology within the Lecture 25 (Functions of the ANS) scope. †

Q44. Regarding autonomic control of salivary gland secretion, which statement is correct?

Correct answer: B. Parasympathetic stimulation produces profuse, watery secretion; sympathetic stimulation produces a small-volume, thick/mucous secretion.

Unlike most glands, salivary glands receive functionally significant dual autonomic innervation: parasympathetic stimulation (via CN VII/IX, acting on M3 receptors) produces abundant, watery, enzyme-rich secretion, while sympathetic stimulation (alpha-adrenergic) produces a smaller volume of thick, mucous-rich secretion — the physiological basis of “dry mouth” during acute sympathetic stress.

Common trap: Option E is tempting since parasympathetic control is the dominant, classically-emphasized pathway, but the sympathetic system does still meaningfully modulate secretion quality — salivary glands are not purely parasympathetic like some other viscera.

Original — standard ANS physiology within the Lecture 25 (Functions of the ANS) scope. †

Lecture 26 — EEG & Sleep Physiology — 1.0 hr — 5 questions

Q45. Which brainwave pattern occurs during deep sleep (Stage N3)?

Correct answer: E. Delta

Deep sleep (Stage N3, slow-wave sleep) is characterized by delta waves — the slowest, highest-amplitude EEG band.

Common trap: Gamma (A) is the FASTEST frequency band, associated with high alertness and active cognitive processing — essentially the opposite end of the spectrum from deep sleep.

Verified — corrected past-paper key (Q150)

Q46. The brain waves seen in a normal, awake, alert state with the eyes open are predominantly:

Correct answer: B. Beta

Alert wakefulness with the eyes open specifically shows beta rhythm; alpha rhythm, by contrast, is associated specifically with relaxed wakefulness with the EYES CLOSED, and “alpha blocking” shifts the pattern to beta the moment the eyes open.

Common trap: Alpha (C) is the most common wrong answer here — it’s the “default” rhythm students associate with normal wakefulness in general, without registering that the eyes-open detail specifically triggers the switch to beta.

Reclassified — corrected past-paper key (Q202), originally in the Mixed/Other Topics bucket; reassigned to Lecture 26 (EEG & Sleep) to match its clear topical scope.

Q47. Which EEG rhythm (4–7 Hz) is characteristically seen during drowsiness and light (Stage 1) sleep in adults, and is also a normal, prominent background rhythm during wakefulness in young children?

Correct answer: B. Theta

Theta rhythm (4–7 Hz) appears during the drowsy transition into sleep (Stage 1 NREM) in adults, and is a normal, prominent background rhythm in awake young children — though it is abnormal if seen diffusely in an awake adult.

Common trap: Delta (A) is the slowest band, tempting because it's also "sleep-associated," but delta belongs to deep Stage 3/4 sleep, not the lighter drowsy transition described here — an easy frequency-band mix-up.

Original — Lecture 26 (EEG & Sleep Physiology) content. †

Q48. Sleep spindles and K-complexes, the EEG hallmarks used to score Stage 2 NREM sleep, arise primarily from oscillatory activity in the:

Correct answer: C. Thalamus (thalamocortical circuitry)

Sleep spindles arise from oscillatory interplay between the thalamic reticular nucleus and thalamocortical relay neurons, projected diffusely to cortex; together with K-complexes (also strongly thalamocortical/cortical in origin), they define Stage 2 NREM sleep on EEG.

Common trap: Locus coeruleus (B) is the brain's main arousal/noradrenergic hub — but it is largely silenced during NREM sleep and plays no role in generating spindles, making it an easy but incorrect "arousal-sounding" distractor.

Original — Lecture 26 (EEG & Sleep Physiology) content. †

Q49. During REM sleep, near-total skeletal muscle paralysis (atonia) is actively produced mainly through inhibition of spinal motor neurons originating from the:

Correct answer: B. Pontine/subcoeruleus region (via glycine/GABA-mediated inhibition)

REM atonia is actively generated by neurons in the pontine (subcoeruleus/peri-locus coeruleus) region, which project to the spinal cord and hyperpolarize alpha motor neurons via glycine and GABA — preventing the sleeper from physically acting out dream content. Failure of this circuit produces REM sleep behavior disorder.

Common trap: Locus coeruleus itself (A) is tempting given its similar name and nearby location, but it is actually SILENCED during REM sleep (it's a wake/NREM-active noradrenergic nucleus) — the true atonia-generating region is the neighboring subcoeruleus area, not the locus coeruleus proper.

Original — Lecture 26 (EEG & Sleep Physiology) content. †

Lecture 27 — Reflex Mechanism — 1.0 hr — 5 questions

Q50. The sensory receptor for the corneal reflex is the:

Correct answer: C. Free nerve endings

The cornea is densely innervated by free nerve endings (via CN V1, the ophthalmic division) and lacks specialized encapsulated mechanoreceptors entirely.

Common trap: Ruffini endings (B) are a real encapsulated mechanoreceptor type used elsewhere in the body, which can make it feel like a legitimate answer — but the cornea specifically contains no encapsulated receptors of any kind.

Verified — corrected past-paper key (Q154)

Q51. What is the maximum possible score on the Glasgow Coma Scale (GCS)?

Correct answer: C. 15

The maximum GCS score is 15, made up of the maximum possible sub-scores: 4 for eye opening, 5 for verbal response, and 6 for motor response (4+5+6=15).

Common trap: “18” (D) is a common arithmetic slip if a student mistakenly assumes each of the three domains maxes out at 6 (6×3=18) instead of remembering the actual 4/5/6 breakdown.

Reclassified — corrected past-paper key (Q200), originally in the Mixed/Other Topics bucket; reassigned to Lecture 27 (Reflex Mechanism), which is where the GCS component breakdown itself is taught in this course.

Q52. On the standard 0–4 scale for grading deep tendon reflexes, a reflex that is present but diminished compared to normal is graded:

Correct answer: B. Grade 1

The standard scale runs: 0 = absent (areflexia), 1 = present but diminished (hyporeflexia), 2 = normal, 3 = increased but not necessarily pathologic (brisk), 4 = markedly hyperactive, pathologic, often with clonus.

Common trap: Grade 3 (D) is often mistakenly assumed to already be “abnormal,” but it is explicitly defined as brisk without necessarily being pathologic — it is Grade 4, with sustained clonus, that is unambiguously pathologic.

Original — Neuro Exam (EP2) reflex-grading content. †

Q53. A patient shows absent superficial abdominal reflexes together with exaggerated (brisk) deep tendon reflexes in the lower limbs. This dissociation is most suggestive of a lesion in the:

Correct answer: B. Corticospinal tract

The superficial abdominal reflex depends on an intact corticospinal pathway to remain present; its loss together with hyperreflexic deep tendon reflexes is a classic sign of a corticospinal (UMN) tract lesion, even though the local segmental reflex arc (T7–T10 upper, T10–L1 lower) is structurally intact.

Common trap: It's tempting to reflexively read “absent reflex” as pointing toward a peripheral (LMN) lesion, but the KEY here is the combination: an absent superficial reflex paired with hyperactive deep reflexes together point to UMN, not peripheral, pathology.

Original — Neuro Exam (EP2) superficial-reflex content. †

Q54. Flicking the distal phalanx of the middle finger and observing reflexive flexion of the thumb and index finger is the Hoffmann sign, and its presence (particularly if asymmetric or accompanied by other findings) is considered:

Correct answer: B. A pathological reflex suggesting a cervical corticospinal (UMN) lesion

The Hoffmann sign is a pathological reflex, functionally analogous to the Babinski sign but for the upper extremity, suggesting a corticospinal tract (UMN) lesion, most often at or above the cervical spinal cord level.

Common trap: It is true that a very mild, symmetric Hoffmann sign is occasionally seen in some healthy, anxious individuals, but as tested here — a genuine, clinically significant pathological reflex — the expected answer is a UMN lesion; don't default to “always normal” just because a benign variant exists.

Original — Neuro Exam (EP2) pathological-reflex content. †

Lecture 28 — Blood Supply of Brain & Meninges — 1.5 hr — 8 questions

Q55. Cerebrospinal fluid from the ventricular system first communicates with the subarachnoid space via which structure?

Correct answer: A. 4th ventricle

CSF exits the ventricular system into the subarachnoid space through the paired lateral foramina of Luschka and the midline foramen of Magendie, all located at the level of the 4th ventricle.

Common trap: Subarachnoid (arachnoid) granulations (C) are where CSF is REABSORBED from the subarachnoid space into the venous blood — an outflow/reabsorption point, not the entry point from the ventricles into the subarachnoid space that the question is asking about.

Verified, resolved — corrected past-paper key (Q160). The original archive extraction left this item ambiguous; resolved here to the foramina at the 4th ventricle, which are the actual anatomical connection between the ventricular system and subarachnoid space.

Q56. A 55-year-old male, post-stroke, is later found on autopsy to have a shaded lesion in the lower pons. The embolus responsible is most likely in which artery?

Correct answer: E. Basilar artery

A lower pons lesion is consistent with basilar artery territory — the basilar artery is the parent vessel supplying the entire pons through its paramedian and circumferential branches.

Common trap: AICA (D) is a tempting distractor since it is also a genuine pontine-supplying vessel, but it supplies only the lateral (dorsolateral) pons, not the full cross-sectional lesion implied by a general “lower pons” shaded lesion — the parent basilar artery is the better fit for a lesion of this scale.

Verified — corrected past-paper key (Q161)

Q57. Which of the following statements about stroke is TRUE?

Correct answer: E. Stroke is neurological injury from occlusion or rupture of cerebral vessels

“Stroke = neurological injury from occlusion or rupture of cerebral vessels” is the correct, standard umbrella definition covering both ischemic (occlusion) and hemorrhagic (rupture) stroke.

Common trap: Option B directly swaps the mechanisms — ischemic stroke results from OCCLUSION, not rupture; rupture is what defines hemorrhagic stroke instead — a very easy pair to reverse under exam pressure.

Verified — corrected past-paper key (Q167)

Q58. Regarding cerebral cortex blood supply, which statement is true?

Correct answer: B. ACA: contralateral leg and micturition

The ACA supplies the medial cortical surface, including the leg/foot motor-sensory homunculus and the micturition control area (paracentral lobule) — the one clearly accurate statement among these options.

Common trap: Option A sounds correct at first glance because MCA genuinely does supply the arm and speech areas, but MCA does NOT significantly supply the leg (that territory belongs to the ACA), making the full statement false.

Verified — corrected past-paper key (Q163)

Q59. Which of the following statements about stroke is FALSE?

Correct answer: B. A TIA (transient ischemic attack) often lasts 2–3 weeks

By definition, a TIA resolves within 24 hours (often within minutes) and causes no permanent infarction on imaging — it does not last 2–3 weeks, making this statement false and the correct answer to “which is false.”

Common trap: *It's tempting to overthink the other options since stroke terminology has many easily-confused pairs (ischemic vs. hemorrhagic mechanisms), but options A, C, D, and E are all standard, accurate definitions — the TIA duration statement is the one clear factual error.*

Verified, resolved — corrected past-paper key (Q165), rephrased for clarity. The original archive extraction flagged multiple options as potentially false; resolved here to the single most specific and clearly false statement (TIA duration).

Q60. A 71-year-old man has a large left MCA territory stroke. An embolus from which vessel is the most likely underlying source?

Correct answer: D. ICA

Large MCA-territory strokes are frequently embolic FROM ipsilateral internal carotid artery (ICA) atherosclerotic disease — a classic artery-to-artery embolism mechanism, since the MCA is the direct continuation of ICA flow.

Common trap: *“MCA” (B) itself is tempting since that's where the stroke occurred, but the question specifically asks for the upstream embolic SOURCE, not the site of the final occlusion — the ICA is the classic proximal source vessel.*

Verified — corrected past-paper key (Q178)

Q61. Which statement about intracranial venous drainage is correct?

Correct answer: B. Superior cerebral veins → superior sagittal sinus

Superior cerebral veins drain directly into the superior sagittal sinus — the one accurate drainage pathway among these options.

Common trap: *Options involving the “great cerebral vein” (A, D) reverse or misdirect its actual connection — the great cerebral vein (of Galen) drains into the straight sinus, not the superior sagittal sinus, in either direction.*

Verified — corrected past-paper key (Q183)

Q62. A patient has an embolus originating from the internal carotid artery. In which branch is the embolus most likely to lodge?

Correct answer: A. MCA

The MCA is the more direct, higher-flow continuation of internal carotid artery blood flow, making it the most common destination for emboli arising from ICA (e.g., carotid plaque) disease.

Common trap: *“VA” (vertebral artery, option E) is a plausible-sounding “major artery” distractor, but the vertebral arteries belong to the separate posterior (vertebrobasilar) circulation and do not receive flow from the ICA at all.*

Verified — corrected past-paper key (Q188)

Lecture 29 — Clinical Correlations — 3.0 hr — 15 questions

Q63. A 40-year-old man has presbyopia (difficulty reading fine print up close). Which statement about the near-focusing (accommodation) reflex is correct?

Correct answer: C. M3 receptors mediate contraction of the ciliary muscle.

The accommodation reflex is a parasympathetic response: M3 muscarinic receptors (via CN III, the Edinger-Westphal pathway) mediate contraction of the ciliary muscle, which paradoxically RELAXES the suspensory (zonular) ligaments, allowing the lens to become rounder for near focus.

Common trap: Option E sounds plausible because ciliary muscle contraction is indeed the trigger, but it reverses the mechanical result — contraction of the ciliary muscle DECREASES, not increases, suspensory ligament tension, which is exactly what allows the lens to round up.

Reclassified — corrected past-paper key (Q193), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as an integrative clinical/autonomic-exam vignette.

Q64. A patient with chorea, basal ganglia atrophy on imaging, and a family history of similar symptoms is suspected of having Huntington's disease. What is the best diagnostic test to confirm this?

Correct answer: C. Genetic testing

Huntington's disease is a single-gene disorder (CAG-repeat expansion in the HTT gene), so genetic testing provides a definitive diagnosis, unlike imaging (which shows nonspecific atrophy) or other supportive-but-nonspecific tests.

Common trap: MRI (B) can show caudate atrophy supporting the clinical picture, but imaging findings are never definitive on their own — the question asks for the BEST diagnostic test, which must be the one that actually confirms the underlying genetic cause.

Reclassified — corrected past-paper key (Q197), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as an integrative diagnostic-workup vignette.

Q65. Which sympathetic receptor is responsible for contracting the internal urinary sphincter, supporting continence during bladder filling?

Correct answer: A. Alpha-1

Sympathetic stimulation contracts the internal urethral sphincter via alpha-1 receptors, supporting continence while the bladder fills, while beta-3 receptors on the detrusor muscle relax the bladder wall itself as a complementary storage mechanism.

Common trap: Beta-3 (B) is a real and relevant bladder receptor, but it acts on the DETRUSOR muscle (relaxing it), not the sphincter — an easy “right receptor, wrong structure” mix-up.

Reclassified — corrected past-paper key (Q199), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as an integrative autonomic-exam vignette.

Q66. What type of fiber provides the major excitatory input to the neostriatum (caudate and putamen)?

Correct answer: B. Corticostriatal, excitatory via Glutamate

The corticostriatal pathway (cortex → striatum) is the major excitatory, glutamatergic input driving basal ganglia circuit activity.

Common trap: The nigrostriatal pathway (A, C) is real and important, but it is dopaminergic, not GABAergic or glutamatergic — both nigrostriatal options here misstate its actual neurotransmitter.

Reclassified — corrected past-paper key (Q205), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as integrative basal-ganglia/clinical content.

Q67. The ganglia housing the first-order (peripheral) taste neurons are found in which of the following structures?

Correct answer: A. Geniculate, Petrosal, Nodose ganglia

The geniculate ganglion (CN VII), petrosal ganglion (CN IX), and nodose ganglion (CN X) house the first-order taste neurons carrying gustatory information from the anterior tongue, posterior tongue, and epiglottis/pharynx respectively.

Common trap: *The spiral ganglion (E) is tempting since it's a real cranial nerve ganglion, but it belongs to the cochlear (hearing) nerve, not to any of the taste-carrying cranial nerves.*

Reclassified — corrected past-paper key (Q209), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as integrative cranial-nerve/exam content.

Q68. In the Gate-control theory of pain, opioid peptides such as enkephalins reduce nociceptive transmission primarily by:

Correct answer: C. Inhibiting Ca²⁺ influx and opening K⁺ channels

Enkephalins act at opioid receptors to inhibit presynaptic Ca²⁺ influx (reducing neurotransmitter release) AND open postsynaptic K⁺ channels (hyperpolarizing the neuron), together reducing substance P release and dampening ascending pain transmission at the dorsal horn.

Common trap: *"Enhancing substance P release" (E) describes exactly the opposite of what enkephalins actually do — they SUPPRESS substance P release as their key analgesic mechanism.*

Reclassified — corrected past-paper key (Q212), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as integrative clinical-pharmacology content relevant to pain management.

Q69. A patient with myasthenia gravis is being managed medically. Which of the following is NOT an appropriate treatment for myasthenia gravis?

Correct answer: E. Dopamine agonists

Dopamine agonists have no role in myasthenia gravis treatment — they are a Parkinson's disease therapy instead. Anticholinesterases, immunosuppression (prednisone), plasmapheresis, and thymectomy are all genuine, standard MG treatments.

Common trap: *Thymectomy (D) might seem like the "odd one out" since it's a surgical option rather than a drug, but it is in fact a well-established MG treatment, especially with thymic pathology — the genuinely unrelated option is the dopaminergic drug class.*

Reclassified — corrected past-paper key (Q216), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as a direct companion question to the myasthenia gravis content already central to this lecture.

Q70. A lesion of the left Meyer's loop (the temporal lobe portion of the optic radiation) results in:

Correct answer: B. Right superior quadrantanopia

Meyer's loop carries fibers representing the contralateral SUPERIOR visual field as it sweeps through the temporal lobe; a left Meyer's loop lesion therefore causes a right superior quadrantanopia (the classic "pie in the sky" defect).

Common trap: *"Right inferior quadrantanopia" (C) describes the mirror-image defect caused by damage to the PARIETAL (dorsal) optic radiation instead — an easy superior/inferior field mix-up between the two optic radiation pathways.*

Reclassified — corrected past-paper key (Q218), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as an integrative visual-field localization vignette.

Q71. Which cranial nerve does NOT contribute afferent fibers to the spinal trigeminal nucleus?

Correct answer: E. CN XI

CN XI (accessory nerve) is purely motor and contributes no sensory afferents to the spinal trigeminal nucleus; by contrast, CN V (its primary input), along with smaller general somatic afferent contributions from CN VII, IX, and X, all converge there.

Common trap: CN VII (B) is a common wrong guess since it feels like a “purely motor” facial nerve, but it does in fact carry a small general somatic afferent component (from the external auditory canal/pinna) into the spinal trigeminal nucleus — the truly all-motor, non-contributing nerve here is CN XI.

Reclassified — corrected past-paper key (Q222), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 as integrative cranial-nerve/exam content.

Q72. A lesion of the left dorsal (parietal) optic radiation produces:

Correct answer: B. Right inferior quadrantanopia

Parietal (dorsal) optic radiation fibers carry the contralateral INFERIOR visual field; a left parietal radiation lesion therefore causes a right inferior quadrantanopia (the classic “pie on the floor” defect) — the mirror image of the superior-field defect caused by a Meyer's loop (temporal) lesion.

Common trap: “Right superior quadrantanopia” (A) describes the Meyer's loop/temporal pathway defect instead — pairing this question with the Meyer's loop question above is a good way to lock in the superior/inferior, temporal/parietal distinction.

Reclassified — corrected past-paper key (Q224), originally in the Mixed/Other Topics bucket; reassigned to Lecture 29 to pair directly with the Meyer's loop vignette above.

Q73. A patient sustains a stab wound causing hemisection of the spinal cord (Brown-Séquard syndrome). On the side of the injury, motor function and vibration/proprioception are lost; on the opposite side, pain and temperature sensation are lost below the level of injury. This lateralization pattern occurs because:

Correct answer: A. The corticospinal and dorsal column tracts have not yet crossed at the level of injury, while the spinothalamic tract already crossed near its segmental entry.

The lateral corticospinal tract and dorsal columns carry ipsilateral information that has already crossed at the medulla (corticospinal) or will cross only much higher up (dorsal columns, at the medulla), so a cord hemisection affects them on the SAME side as the lesion; the spinothalamic tract, by contrast, crosses within one or two segments of entering the cord, so its fibers are already on the OPPOSITE side by the time they ascend past the lesion level.

Common trap: It's easy to expect every deficit to land on the same side as the lesion, but the spinothalamic tract's early segmental decussation is exactly why pain/temperature loss shows up contralateral to the lesion while everything else (motor, vibration, proprioception) remains ipsilateral.

Original — Neuro Exam (EP2) spinal cord syndrome content. †

Q74. An elderly man falls and sustains a hyperextension neck injury. He develops weakness that is much greater in the arms than the legs, with a 'cape-like' loss of pain and temperature sensation across the shoulders and upper arms, while leg strength and lower-body sensation are relatively preserved. This is most consistent with:

Correct answer: C. Central cord syndrome

Central cord syndrome, classically from hyperextension injury in an already-degenerated (spondylotic) cervical spine in an older patient, damages the centrally-located cervical fibers first — these carry the arm-predominant corticospinal and spinothalamic fibers — producing arm-greater-than-leg weakness and a cape-like sensory loss, while the more peripherally-located leg fibers are relatively spared.

Common trap: Anterior cord syndrome (B) also causes bilateral motor and pain/temperature loss, but it does not selectively spare the legs over the arms — the classic arm-greater-than-leg pattern with a “cape” sensory distribution specifically points to central, not anterior, cord involvement.

Original — Neuro Exam (EP2) spinal cord syndrome content. †

Q75. A patient with a chronic, length-dependent polyneuropathy shows sensory loss and weakness that is noticeably worse distally (feet) than proximally. This distal-predominant pattern is most typical of which underlying process?

Correct answer: B. Axonal polyneuropathy

Axonal polyneuropathies are classically length-dependent, affecting the longest axons first and therefore producing a “stocking-glove,” distal-greater-than-proximal pattern of weakness and sensory loss.

Common trap: It's easy to assume all polyneuropathies are simply “distal,” but demyelinating processes (e.g., Guillain-Barré syndrome, CIDP) specifically break this rule by also significantly involving proximal segments, producing weakness that is roughly proximal ≈ distal — length-dependence is the axonal signature, not a universal polyneuropathy rule.

Original — Neuro Exam (EP2) peripheral nerve pattern-of-neuropathy content. †

Q76. When testing vibration sense with a 128 Hz tuning fork, which of the following describes the correct technique?

Correct answer: B. Place the struck, vibrating fork on a bony prominence (e.g., the great toe) and ask the patient to report when the vibration stops.

Correct technique places the struck, vibrating tuning fork on a distal bony prominence (classically the dorsum of the great toe interphalangeal joint) and asks the patient to report the moment the vibration stops; testing starts distally because that is the first place vibration sense is lost in a length-dependent neuropathy.

Common trap: Simply asking ‘do you feel that?’ (A) only confirms the patient perceives touch/pressure from the fork's placement — it does not test true vibration sense and misses subtle deficits, which is why this is explicitly the wrong technique to use.

Original — Neuro Exam (EP2) sensory examination content. †

Q77. The NIH Stroke Scale (NIHSS), used at the bedside in acute stroke care, is used primarily to:

Correct answer: B. Quantify stroke severity and help guide/monitor treatment such as thrombolysis

The NIHSS is a standardized bedside scale that quantifies acute stroke severity (via items covering consciousness, gaze, visual fields, motor and sensory function, language, and neglect); it is used both to guide decisions such as thrombolysis (rt-PA) eligibility and to track a patient's neurological status over time, e.g., before and after treatment.

Common trap: NIHSS is a clinical severity score, not an imaging or vascular-territory-localizing tool — a low or high score by itself does not tell you which artery is occluded, only how severe the neurological deficit currently is.

Original — Neuro Exam (EP2) stroke-case content. †
